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Inventor(s)	Wu; Yuan-Lin et al.

Method for manufacturing display device

Abstract

The disclosure provides a method for manufacturing a display device, including the following steps. A substrate is provided. A pixel circuit is formed on the substrate. A light-emitting unit is formed on the pixel circuit. A touch sensing unit is formed on the light-emitting unit. An insulating layer is formed on the touch sensing unit. An anti-reflection layer is formed on the insulating layer. The method for manufacturing the display device of the embodiment of the disclosure can improve reliability.

Inventors: Wu; Yuan-Lin (Miao-Li County, TW), Lius; Chandra (Miao-Li County, TW), Tsai; Tsung-Han (Miao-Li County, TW)

Applicant: Innolux Corporation (Miao-Li County, TW)

Family ID: 80447628

Assignee: Innolux Corporation (Miaoli County, TW)

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Primary Examiner: Booth; Richard A

Attorney, Agent or Firm: JCIPRNET

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims the priority benefit of China application serial no. 202110255593.6, filed on Mar. 9, 2021. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Technical Field

(2) The disclosure relates to a method for manufacturing a display device, and more particularly, to a method for manufacturing a display device which can improve reliability.

Description of Related Art

(3) Display devices have been widely applied to electronic devices such as mobile phones, televisions, monitors, tablet computers, car displays, wearable devices, and desktop computers. With the vigorous development of electronic products, the requirements for the display quality of the display devices also increase, such that the display devices are constantly improving towards display effects of high brightness, low energy consumption, high resolution, or high saturation.

Meanwhile, the method for manufacturing the display device is constantly improving towards reduced process time, reduced process steps, or an improved process sequence.

SUMMARY

(4) The disclosure provides a method for manufacturing a display device which can improve the reliability of a display device such as improving the bending endurance of the display device, improving the service life of the display device, or improving both of the above, but the disclosure is not limited thereto.

(5) The disclosure provides a method for manufacturing a display device, including the following steps. A substrate is provided. A pixel circuit is formed on the substrate. A light-emitting unit is formed on the pixel circuit. A touch sensing unit is formed on the light-emitting unit. An insulating layer is formed on the touch sensing unit. An anti-reflection layer is formed on the insulating layer.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings are included to provide a further understanding of the disclosure, and are incorporated in and constitute a part of this specification. The drawings illustrate embodiments of the disclosure and, together with the description, serve to explain the principles of the disclosure.

(2) FIG. 1 is a schematic partial cross-sectional view of a display device according to some embodiments of the disclosure.

(3) FIG. 2 is a flowchart of a method for manufacturing a display device according to an embodiment of the disclosure.

(4) FIG. 3A to FIG. 3H are cross-sectional views showing a process of a method for manufacturing a display device according to some embodiments of the disclosure.

(5) FIG. 3H' and FIG. 3H'' are partial cross-sectional views of a display device according to some embodiments of the disclosure.

(6) FIG. 4 is a cross-sectional view showing a process of a method for manufacturing a display device according to another embodiment of the disclosure.

(7) FIG. 5A and FIG. 5B are cross-sectional views showing a process of a method for manufacturing a display device according to another embodiment of the disclosure.

(8) FIG. 6A and FIG. 6B are respectively cross-sectional views showing a process of a method for manufacturing a display device according to some embodiments of the disclosure.

(9) FIG. 7 is a cross-sectional view showing a process of a method for manufacturing a display device according to another embodiment of the disclosure.

(10) FIG. 8A and FIG. 8B are schematic partial cross-sectional views of a display device according to some embodiments of the disclosure.

(11) FIG. 9 is a schematic top view of a display device according to another embodiment of the disclosure.

(12) FIG. 10A to FIG. 10E are cross-sectional views showing a process of a method for manufacturing the display device of FIG. 9 taken along section line A-A'.

(13) FIG. 11A to FIG. 11C are cross-sectional views showing a process of a method for manufacturing the display device of FIG. 9 taken along section line B-B'.

(14) FIG. 12A to FIG. 12C are cross-sectional views showing a process of a method for manufacturing the display device of FIG. 9 taken along section line A-A' according to another embodiment.

(15) FIG. 13 is a flowchart of a method for manufacturing a display device according to another embodiment of the disclosure.

(16) FIG. 14A to FIG. 14D are cross-sectional views showing a process of a method for

manufacturing a display device according to another embodiment of the disclosure.

(17) FIG. 15 is a flowchart of a method for manufacturing a display device according to another embodiment of the disclosure.

(18) FIG. 16A to FIG. 16D are cross-sectional views showing a process of a method for manufacturing a display device according to another embodiment of the disclosure.

DESCRIPTION OF THE EMBODIMENTS

(19) The disclosure may be understood by referring to the following detailed description with reference to the accompanying drawings. It is noted that for comprehension of the reader and simplicity of the drawings, in the drawings of the disclosure, only a part of the electronic device is shown, and specific components in the drawings are not necessarily drawn to scale. Moreover, the quantity and the size of each component in the drawings are only schematic and are not intended to limit the scope of the disclosure.

(20) In the following specification and claims, the terms “having”, “including”, “comprising” etc. are open-ended terms, so they should be interpreted to mean “including but not limited to . . .”.

(21) It should be understood that when a component or a film layer is described as being “on” or “connected to” another component or film layer, it may be directly on or connected to the another component or film layer, or there is an intervening component or film layer therebetween (i.e., being indirectly on or indirectly connected). Conversely, when a component or film layer is described as being “directly on” or “directly connected to” another component or film layer, there is no intervening component or film layer therebetween.

(22) The terms such as “first”, “second”, “third”, etc. may be used to describe components, but the components should not be limited by these terms. The terms are only intended to distinguish a component from another component in the specification. It is possible that the claims do not use the same terms and replace the terms with “first”, “second”, “third” etc. according to the sequence declared in the claims. Accordingly, in the specification, a first component may be described as a second component in the claims.

(23) In some embodiments of the disclosure, unless specifically defined, terms related to bonding and connection such as “connect”, “interconnect”, etc. may mean that two structures are in direct contact, or that two structures are not in direct contact and another structure is provided therebetween. The terms related to bonding and connection may also cover cases where two structures are both movable or two structures are both fixed. In addition, the term “couple” includes any direct and indirect electrical connection means.

(24) In the disclosure, a length, width, thickness, height, or area, or a distance or spacing between components may be measured by optical microscopy (OM), a scanning electron microscope (SEM), a film thickness profile measuring instrument (α -step), an ellipsometer, or other suitable methods. Specifically, according to some embodiments, a cross-sectional structure image containing components to be measured may be obtained by a scanning electron microscope to measure the width, thickness, height, or area of each component, or the distance or spacing between the components, but the disclosure is not limited thereto. In addition, there may be a certain error between any two values or directions used for comparison.

(25) In the disclosure, the terms such as “approximately”, “about”, or “substantially” are generally interpreted as one value being within a range of plus or minus 10%, plus or minus 5%, plus or minus 3%, plus or minus 2%, plus or minus 1%, or plus or minus 0.5% of a given value. Herein, a given value refers to an approximate value, i.e., in the absence of a specific expression of “approximately”, “about”, or “substantially”, the meaning of “approximately”, “about”, or “substantially” may still be implied. In addition, the expression “a given range is between a first numerical value and a second numerical value” means that the given range includes the first numerical value, the second numerical value, and other values therebetween.

(26) The electronic device of the disclosure may include a display device, an antenna device (e.g., a liquid crystal antenna), a sensing device, a light-emitting device, a touch device, or a splicing

device, but is not limited thereto. The electronic device may include a bendable or flexible electronic device. The shape of the electronic device may be rectangular, circular, polygonal, a shape with curved edges, or other suitable shapes. The display device may include, for example, a light-emitting diode (LED), a liquid crystal, a fluorescence, a phosphor, a quantum dot (QD), other suitable materials, or a combination of the above, but is not limited thereto. The light-emitting diode may include, for example, an organic light-emitting diode (OLED), an inorganic light-emitting diode, a mini LED, a micro LED, a quantum dot LED (QLED or QDLED), other suitable materials, or any combination of the above, but is not limited thereto. The display device may include, for example, a splicing display device but is not limited thereto. The antenna device may include, for example, a liquid crystal antenna but is not limited thereto. The antenna device may include, for example, an antenna splicing device but is not limited thereto. It is noted that the electronic device may be any combination of the above but is not limited thereto. The electronic device may have peripheral systems such as a driving system, a control system, a light source system, a rack system, etc. to support a display device, an antenna device, or a splicing device. Hereinafter, a display device will be described to illustrate the content of the disclosure, but the disclosure is not limited thereto.

(27) In the disclosure, the features in multiple different embodiments described below may be replaced, combined, and/or mixed to form other embodiments without departing from the spirit of the disclosure. The features of the embodiments may be arbitrarily mixed and combined as long as they do not depart from or conflict with the spirit of the disclosure.

(28) FIG. 1 is a schematic partial cross-sectional view of a display device according to some embodiments of the disclosure. Referring to FIG. 1, a display device **100** of this embodiment includes the following components sequentially from bottom to top: a substrate **110**, a pixel circuit **120**, a light-emitting unit **130**, an encapsulation layer **140**, a touch sensing unit **150**, an insulating layer **160**, an anti-reflection layer **170**, and a cover layer **180**. In this embodiment, the substrate **110** has a thickness **T1**. The thickness **T1** is, for example, the maximum thickness of the substrate **110** measured along the normal direction of the substrate **110**. The substrate **110** may include a rigid substrate, a flexible substrate, or a combination of the above. For example, the material of the substrate **110** may include glass, quartz, sapphire, ceramic, polycarbonate (PC), polyimide (PI), polyethylene terephthalate (PET), other suitable substrate materials, or combinations of the above, but is not limited thereto.

(29) The pixel circuit **120** is disposed on the substrate **110** and includes a transistor **121**. The light-emitting unit **130** is disposed on the pixel circuit **120** and is electrically connected to the transistor **121** in the pixel circuit **120**.

(30) The encapsulation layer **140** is disposed on the light-emitting unit **130**. The encapsulation layer **140** includes a first insulating layer **141**, a planarizing layer **142**, and a second insulating layer **143**. The first insulating layer **141** is disposed on the light-emitting unit **130** and has a thickness **T2**. The planarizing layer **142** is disposed on the first insulating layer **141** and has a thickness **T3**. The second insulating layer **143** is disposed on the planarizing layer **142** and has a thickness **T4**. The thickness **T2** is, for example, the maximum thickness of the first insulating layer **141** measured along the normal direction of the substrate **110**. The thickness **T3** is, for example, the maximum thickness of the planarizing layer **142** measured along the normal direction of the substrate **110**. The thickness **T4** is, for example, the maximum thickness of the second insulating layer **143** measured along the normal direction of the substrate **110**. In this embodiment, the first insulating layer **141**, the planarizing layer **142**, and the second insulating layer **143** may respectively have a single-layer structure or a multi-layer structure. The materials of the first insulating layer **141** and the second insulating layer **143** may include, for example, an inorganic material (e.g., silicon nitride or other suitable inorganic materials) but are not limited thereto. The material of the planarizing layer **142** may include, for example, an organic material but is not limited thereto.

(31) In this embodiment, the thickness **T4** of the second insulating layer **143** is, for example,

greater than the thickness T2 of the first insulating layer **141** (i.e., $T4 > T2$), but the disclosure is not limited thereto. The ratio of the thickness T4 of the second insulating layer **143** to the thickness T2 of the first insulating layer **141** is, for example, greater than 1 and less than or equal to 3 (i.e., $1 < T4/T2 \leq 3$), but the disclosure is not limited thereto. The thickness T3 of the planarizing layer **142** is, for example, greater than the sum of the thickness T2 of the first insulating layer **141** and the thickness T4 of the second insulating layer **143** (i.e., $T3 > T2 + T4$), but the disclosure is not limited thereto. The ratio of the thickness T3 of the planarizing layer **142** to the sum (i.e., the sum of the thickness T2 of the first insulating layer **141** and the thickness T4 of the second insulating layer **143**) is, for example, greater than or equal to 3 and less than or equal to 6 (i.e., $3 \leq T3/(T2 + T4) \leq 6$), but the disclosure is not limited thereto. In addition, in this embodiment, the thickness T2 of the first insulating layer **141** is, for example, greater than or equal to 0.5 μm and less than or equal to 1 μm (i.e., $0.5 \mu\text{m} \leq T2 \leq 1 \mu\text{m}$) but is not limited thereto. The thickness T3 of the planarizing layer **142** is, for example, greater than or equal to 5 μm and less than or equal to 10 μm (i.e., $5 \mu\text{m} \leq T3 \leq 10 \mu\text{m}$) but is not limited thereto. The thickness T4 of the second insulating layer **143** is, for example, greater than or equal to 0.8 μm and less than or equal to 1.5 μm (i.e., $0.8 \mu\text{m} \leq T4 \leq 1.5 \mu\text{m}$) but is not limited thereto.

(32) The touch sensing unit **150** is disposed on the encapsulation layer **140**. The insulating layer **160** is disposed on the touch sensing unit **150**. The anti-reflection layer **170** is disposed on the insulating layer **160** and has a thickness T5. The thickness T5 is, for example, the maximum thickness of the anti-reflection layer **170** measured along the normal direction of the substrate **110**. The cover layer **180** is disposed on the anti-reflection layer **170** and has a thickness T6. The thickness T6 is, for example, the maximum thickness of the cover layer **180** measured along the normal direction of the substrate **110**. In addition, in this embodiment, the thickness T1 of the substrate **110** is, for example, greater than the thickness T5 of the anti-reflection layer **170** but is not limited thereto. The thickness T6 of the cover layer **180** is, for example, greater than the thickness T5 of the anti-reflection layer **170** but is not limited thereto.

(33) Furthermore, the display device **100** of this embodiment further includes a buffer layer **111**. The buffer layer **111** is disposed between the pixel circuit **120** and the substrate **110**. The buffer layer **111** may have a single-layer structure or a multi-layer structure, and the buffer layer **111** may include an organic material, an inorganic material, or a combination of the above but is not limited thereto.

(34) Other embodiments will be provided and described below. It is noted herein that the reference numerals and part of the descriptions of the above embodiment apply to the following embodiments, where the same numerals are used to represent the same or similar components, and descriptions of the same technical contents are omitted. Reference may be made to the above embodiment for the descriptions of the omitted contents, which will not be repeated in the following embodiments.

(35) FIG. 2 is a flowchart of a method for manufacturing a display device according to an embodiment of the disclosure. FIG. 3A to FIG. 3H are cross-sectional views showing a process of a method for manufacturing a display device according to some embodiments of the disclosure. FIG. 3H' and FIG. 3H'' are partial cross-sectional views of a display device according to some embodiments of the disclosure.

(36) First, referring to FIG. 2 and FIG. 3A at the same time, in a method for manufacturing a display device **100a** of this embodiment, first, step S200 and step S210 are performed to provide a substrate **110** and form a pixel circuit **120** on the substrate **110**. Specifically, the pixel circuit **120** may include a transistor **121**, a gate insulating layer GI, an insulating layer **122**, and an insulating layer **123**. The transistor **121** may include a gate electrode GE, a portion of the gate insulating layer GI, a source electrode SD1, a drain electrode SD2, and a semiconductor SE. The semiconductor SE is disposed on the substrate **110**, the gate insulating layer GI is disposed on the semiconductor SE, the gate electrode GE is disposed on the gate insulating layer GI, the insulating layer **122** is

disposed on the gate electrode GE, the source electrode SD1 and the drain electrode SD2 are disposed on the insulating layer 122, and the insulating layer 123 is disposed on the source electrode SD1 and the drain electrode SD2. In this embodiment, the material of the semiconductor SE may include an amorphous silicon, a low temperature polysilicon (LIPS), a metal oxide (e.g., indium gallium zinc oxide (IGZO)), other suitable materials, or a combination of the above but is not limited thereto. In other embodiments, different transistors may include different semiconductor layer materials, but the disclosure is not limited thereto. Moreover, although the transistor 121 of this embodiment is a top-gate transistor, the disclosure is not limited thereto. In other embodiments, the transistor may also include a bottom-gate transistor, a top-gate transistor, and/or a double-gate transistor, but the disclosure is not limited thereto. In addition, the display device 100a of this embodiment further includes a buffer layer 111, and the buffer layer 111 is disposed between the gate insulating layer GI of the pixel circuit 120 and the substrate 110.

(37) Then, referring to FIG. 2 and FIG. 3B at the same time, step S220 is performed to form a light-emitting unit 130 on the pixel circuit 120. Specifically, the light-emitting unit 130 includes a first electrode 131, an insulating layer 132, a light-emitting element 133, and a second electrode 134. The first electrode 131 is disposed on the insulating layer 123. The insulating layer 132 is disposed on the first electrode 131. The insulating layer 132 has an opening 132a to expose part of the first electrode 131. The light-emitting element 133 is disposed in the opening 132a and on the first electrode 131 to electrically connect to the drain electrode SD2 of the transistor 121 via the first electrode 131. The second electrode 134 is disposed on the insulating layer 132 and in the opening 132a and is electrically connected to the light-emitting element 133. In this embodiment, a light-emitting element 133 (e.g., an organic light-emitting diode) may be formed (e.g., by deposition, but the disclosure is not limited thereto) directly on the first electrode 131, or a light-emitting element 133 (e.g., a micro light-emitting diode) which has been completed may be placed on the first electrode 131 by indirect transfer. The light-emitting element 133 may include light-emitting diodes (LED) of different colors, e.g., a red light-emitting diode 133R, a green light-emitting diode 133G, and a blue light-emitting diode 133B, but is not limited thereto.

(38) Next, referring to FIG. 2 and FIG. 3C at the same time, step S230 is performed to form an encapsulation layer 140 on the light-emitting unit 130. Specifically, the encapsulation layer 140 has a multi-layer structure, including a first insulating layer 141, a planarizing layer 142, and a second insulating layer 143. The first insulating layer 141 is disposed on the second electrode 134 of the light-emitting unit 130. The planarizing layer 142 is disposed on the first insulating layer 141. The second insulating layer 143 is disposed on the planarizing layer 142. The first insulating layer 141 and the second insulating layer 143 are respectively located on two opposite sides of the planarizing layer 142. In addition, the arrangement of the planarizing layer 142 may provide a substantially flat surface. Specifically, for example, thicknesses may be measured along the normal direction of the substrate 110 from the surface of the planarizing layer 142 to the surface of the substrate 110, and when the difference between any two thicknesses is less than or equal to 1 μm , the planarizing layer 142 may be regarded as flat, so that the subsequently manufactured layer (e.g., an anti-reflection layer 170) may have a better effect.

(39) Then, referring to FIG. 2 and FIG. 3D at the same time, step S240 and step S250 are performed to form a touch sensing unit 150 on the light-emitting unit 130 and form an insulating layer 160 on the touch sensing unit 150. Specifically, the touch sensing unit 150 may include a first metal layer 151, an insulating layer 152, a conductive via 153, and a second metal layer 154, and the first metal layer 151 and the second metal layer 154 may include a patterned conductive layer, but the disclosure is not limited thereto. The first metal layer 151 is disposed on the second insulating layer 143 of the encapsulation layer 140. The first metal layer 151 is disposed corresponding to a non-light-emitting region 101 and is not disposed corresponding to light-emitting regions 102R, 102G, and 102B. Since the light-emitting element 133 of this embodiment emits light in a direction away from the substrate 110, the light-emitting regions 102R, 102G, and

102B are disposed corresponding to the light-emitting direction of the light-emitting element **133**. In other words, the light-emitting regions **102R**, **102G**, and **102B** may be defined as regions that overlap with the light-emitting element **133** (the red light-emitting diode **133R**, the green light-emitting diode **133G**, and the blue light-emitting diode **133B**) in the light-emitting direction of the light-emitting element **133** (the red light-emitting diode **133R**, the green light-emitting diode **133G**, and the blue light-emitting diode **133B**), and the non-light-emitting region **101** may be defined as a region that does not overlap with the light-emitting element **133** (the red light-emitting diode **133R**, the green light-emitting diode **133G**, and the blue light-emitting diode **133B**) in the light-emitting direction of the light-emitting element **133** (the red light-emitting diode **133R**, the green light-emitting diode **133G**, and the blue light-emitting diode **133B**). The insulating layer **152** is disposed on the first metal layer **151**. The second metal layer **154** is disposed on the insulating layer **152** and is electrically connected to the first metal layer **151** via the conductive via **153** penetrating the insulating layer **152**. In this embodiment, the materials of the first metal layer **151** and the second metal layer **154** may include, for example, a stack of three metal layers, in which the first layer and the third layer are titanium, and the second layer located between the first layer and the third layer is aluminum, but the disclosure is not limited thereto. The insulating layer **152** may have a single-layer structure or a multi-layer structure, and the insulating layer **152** may include an inorganic material (e.g., silicon nitride or other suitable inorganic materials) but is not limited thereto.

(40) In this embodiment, a thickness **T7** of the first metal layer **151** is, for example, greater than or equal to $0.1\ \mu\text{m}$ and less than or equal to $0.4\ \mu\text{m}$ (i.e., $0.1\ \mu\text{m} \leq T7 \leq 0.4\ \mu\text{m}$) but is not limited thereto. A thickness **T8** of the insulating layer **152** is, for example, greater than or equal to $0.1\ \mu\text{m}$ and less than or equal to $0.4\ \mu\text{m}$ (i.e., $0.1\ \mu\text{m} \leq T8 \leq 0.4\ \mu\text{m}$) but is not limited thereto. A thickness **T9** of the second metal layer **154** is, for example, greater than or equal to $0.2\ \mu\text{m}$ and less than or equal to $0.5\ \mu\text{m}$ (i.e., $0.2\ \mu\text{m} \leq T9 \leq 0.5\ \mu\text{m}$) but is not limited thereto. If the thickness **T7** of the first metal layer **151** is less than $0.1\ \mu\text{m}$ or the thickness **T9** of the second metal layer **154** is less than $0.2\ \mu\text{m}$, the impedance will increase and signal transmission will be affected. If the thickness **T7** of the first metal layer **151** is greater than $0.4\ \mu\text{m}$ or the thickness **T9** of the second metal layer **154** is greater than $0.5\ \mu\text{m}$, the topography will be uneven, which will affect the polarization effect of a subsequently manufactured anti-reflection layer **170**. In addition, in this embodiment, the thickness **T9** of the second metal layer **154** is, for example, greater than the thickness **T7** of the first metal layer **151** (i.e., $T9 > T7$) but is not limited thereto. The thickness **T9** of the second metal layer **154** is, for example, greater than the thickness **T8** of the insulating layer **152**, and the thickness **T8** of the insulating layer **152** is, for example, greater than the thickness **T7** of the first metal layer **151** (i.e., $T9 > T8 > T7$), but the disclosure is not limited thereto. The ratio of the thickness **T9** of the second metal layer **154** to the thickness **T7** of the first metal layer **151** is, for example, greater than or equal to 1 and less than or equal to 1.5 (i.e., $1 \leq T9/T7 \leq 1.5$), but the disclosure is not limited thereto.

(41) In this embodiment, the insulating layer **160** is disposed on the insulating layer **152** and the second metal layer **154** of the touch sensing unit **150**. The insulating layer **160** may have a single-layer structure or a multi-layer structure, and the insulating layer **160** may include an organic material, an inorganic material, or a combination of the above but is not limited thereto.

(42) Then, referring to FIG. 2, FIG. 3E, FIG. 3F, and FIG. 3G at the same time, step **S260** and step **S260a** are performed to form an anti-reflection layer **170** on the insulating layer **160**. The anti-reflection layer **170** may include, for example, a patterned anti-reflection layer but is not limited thereto, and the anti-reflection layer **170** may include a color filter layer **171R**, a color filter layer **171G**, and a color filter layer **171B**. In this embodiment, the anti-reflection layer **170** is formed on the insulating layer **160** by, for example, a semiconductor manufacturing process, and is not provided by adhering an anti-reflection layer which has been completed onto the insulating layer **160** through an adhesive layer. In this embodiment, the anti-reflection layer **170** may be formed directly on the insulating layer **160**, so that there is no other layer between the anti-reflection layer **170** and the insulating layer **160**. In this embodiment, the anti-reflection layer **170** may be formed

on the insulating layer **160** by, for example, an ink-jet printing process, but disclosure does not specifically limit the formation method of the anti-reflection layer **170**. Specifically, referring to FIG. **3E**, by an ink-jet printing process, the red color filter layer **171R** is formed on the insulating layer **160** and is disposed corresponding to the light-emitting region **102R** (i.e., the light-emitting region of the red light-emitting diode **133R**) and the non-light-emitting region **101**, but is not disposed corresponding to the light-emitting region **102G** (i.e., the light-emitting region of the green light-emitting diode **133G**) and the light-emitting region **102B** (i.e., the light-emitting region of the blue light-emitting diode **133B**). Next, referring to FIG. **3F**, the green color filter layer **171G** is formed on the insulating layer **160** and the color filter layer **171R** and is disposed corresponding to the light-emitting region **102G** and the non-light-emitting region **101**, but is not disposed corresponding to the light-emitting region **102R** and the light-emitting region **102B**. Next, referring to FIG. **3G**, the blue color filter layer **171B** is formed on the insulating layer **160** and the color filter layer **171G** and is disposed corresponding to the light-emitting region **102B** and the non-light-emitting region **101**, but is not disposed corresponding to the light-emitting region **102R** and the light-emitting region **102G**. The stack of three color filter layers (i.e., the stack of the color filter layer **171R**, the color filter layer **171G**, and the color filter layer **171B**) disposed on the non-light-emitting region **101** may have a light-shielding effect. So far, the anti-reflection layer **170** of this embodiment has been completed.

(43) In some embodiments, the anti-reflection layer **170** may also be formed on the insulating layer **160** by, for example, a coating process, and the anti-reflection layer **170** may also be formed on the insulating layer **160** by, for example, a photolithography process. Specifically, one layer of red color filter layer **171R** is formed on the insulating layer **160** by a coating process, so that the red color filter layer **171R** covers the light-emitting region **102R**, the light-emitting region **102G**, the light-emitting region **102B**, and the non-light-emitting region **101** (not shown). Next, the red color filter layer **171R** is etched by a photolithography process to etch the color filter layer **171R** on the light-emitting region **102G** and the light-emitting region **102B** and expose the insulating layer **160**, as shown in FIG. **3E**. Then, according to steps similar to those described above, the green color filter layer **171G** is formed by a coating process and a photolithography process, so that the green color filter layer **171G** is disposed in the light-emitting region **102G** and the non-light-emitting region **101**, and the color filter layer **171R** of the light-emitting region **102R** and the insulating layer **160** of the light-emitting region **102B** are exposed, as shown in FIG. **3F**. Then, according to steps similar to those described above, the blue color filter layer **171B** is formed by a coating process and a photolithography process, so that the blue color filter layer **171B** is disposed in the light-emitting region **102B** and the non-light-emitting region **101**, and the color filter layer **171R** of the light-emitting region **102R** and the color filter layer **171G** of the light-emitting region **102G** are exposed, as shown in FIG. **3G**.

(44) Although in the method for manufacturing the display device **100a** of this embodiment, the red color filter layer **171R** is formed first, and then the green color filter layer **171G** and the blue color filter layer **171B** are formed, the disclosure does not specifically limit the sequence of forming the color filter layer. In some embodiments, the green color filter layer **171G** or the blue color filter layer **171B** may also be formed first.

(45) Then, referring to FIG. **2** and FIG. **3H** at the same time, step **S270** is performed to form a cover layer **180** on the anti-reflection layer **170**. In this embodiment, the cover layer **180** may be formed on the anti-reflection layer **170** by a coating process, but the disclosure does not specifically limit the formation method of the cover layer **180**. Specifically, the cover layer **180** includes a planarizing layer **181** and a planarizing layer **182**. In this embodiment, by a coating process, the planarizing layer **181** is first formed on the color filter layer **171R** of the light-emitting region **102R**, on the color filter layer **171G** of the light-emitting region **102G**, on the color filter layer **171B** of the light-emitting region **102B**, and on the color filter layer **171B** of the non-light-emitting region **101** to perform planarization. Then, the planarizing layer **182** is formed on the planarizing

layer **181** by a coating process. The planarizing layer **181** and the planarizing layer **182** may have a single-layer structure or a multi-layer structure, and the planarizing layer **181** and the planarizing layer **182** may include an organic material but are not limited thereto. For example, the planarizing layer **181** and the planarizing layer **182** may include polyethylene terephthalate, polyimide, or a combination of the above but are not limited thereto. So far, the display device **100a** of this embodiment has been completed.

(46) In some embodiments, the cover layer **180** may also be formed on the anti-reflection layer **170** by, for example, a coating process and an adhesion process. Referring to FIG. **3H'**, the cover layer **180** includes a planarizing layer **181**, an adhesive layer **183**, and a substrate **184**. Specifically, by a coating process, the planarizing layer **181** is first formed on the color filter layer **171R** of the light-emitting region **102R**, on the color filter layer **171G** of the light-emitting region **102G**, on the color filter layer **171B** of the light-emitting region **102B**, and on the color filter layer **171B** of the non-light-emitting region **101** to perform planarization. Next, the adhesive layer **183** is formed on the planarizing layer **181** by a coating process. Then, the substrate **184** is adhered onto the adhesive layer **183**. The substrate **184** may have a multi-layer structure, and may include a resin layer **1841**, an adhesive layer **1842**, a glass **1843**, and a hard coating layer **1844** but is not limited thereto. In other embodiments, a substrate **184a** may also include a resin layer **1841**, an adhesive layer **1842**, a resin layer **1845**, and a hard coating layer **1844**, as shown in FIG. **3H''**. The adhesive layer **183** and the adhesive layer **1842** may include an optical adhesive (OCA), the material of the resin layer **1841** may include polyethylene terephthalate, the material of the hard coating layer **1844** may include poly(methyl methacrylate) (PMMA), and the material of the resin layer **1845** may include colorless polyimide, but the disclosure is not limited thereto.

(47) In brief, in the method for manufacturing the display device **100a** of the embodiment of the disclosure, compared to a method of attaching an anti-reflection layer externally onto the insulating layer, the method of forming the anti-reflection layer **170** on the insulating layer **160** in the embodiment of the disclosure can reduce the overall thickness, reduce the bending stress, improve the reliability, or increase the service life. For example, when the display device **100a** is a foldable display device or a stretchable display device (as shown in FIG. **9**), and when the display device is to be bent, rolled, or stretched, compared to the method of attaching an anti-reflection layer externally onto the insulating layer, the method of forming the anti-reflection layer **170** on the insulating layer **160** in the embodiment of the disclosure can reduce the bending stress to suppress separation of the anti-reflection layer, which thereby improves the reliability of the display device or increases the service life of the display device.

(48) FIG. **4** is a cross-sectional view showing a process of a method for manufacturing a display device according to another embodiment of the disclosure. FIG. **4** shows a step continuing FIG. **3D** and replacing FIG. **3E** to FIG. **3G**. The same materials or methods may apply to the same or similar components in the embodiment of FIG. **4** and the embodiment of FIG. **3A** to FIG. **3H**, so the same and similar descriptions in the two embodiments will not be repeated herein, and the main difference between the two embodiments will be described.

(49) Referring to FIG. **4**, a black matrix layer **BM** is formed on the insulating layer **160**. The black matrix layer **BM** may include a patterned black matrix layer and is disposed corresponding to the non-light-emitting region **101**. The black matrix layer **BM** may expose the light-emitting region **102R**, the light-emitting region **102G**, and the light-emitting region **102B** on the insulating layer **160**. Then, by an ink-jet printing process, a red color filter layer **171R** is formed on the insulating layer **160** and is disposed corresponding to the light-emitting region **102R**, a green color filter layer **171G** is formed on the insulating layer **160** and is disposed corresponding to the light-emitting region **102G**, and a blue color filter layer **171B** is formed on the insulating layer **160** and is disposed corresponding to the light-emitting region **102B**. Therefore, in this embodiment, the black matrix layer **BM** replaces the stack of three color filter layers located in the non-light-emitting region **101** in FIG. **3G** and may have the effect of shielding light or as a blocking wall. The

expression “corresponding to” described in the disclosure may mean, for example, that at least part of a component A and at least part of a component B overlap with each other in the normal direction of the substrate **110**, or the component A and the component B completely overlap with each other.

(50) FIG. 5A and FIG. 5B are cross-sectional views showing a process of a method for manufacturing a display device according to another embodiment of the disclosure. FIG. 5A and FIG. 5B show steps continuing FIG. 3D and replacing FIG. 3E to FIG. 3G. The same materials or methods may apply to the same or similar components in the embodiment of FIG. 5A to FIG. 5B and the embodiment of FIG. 3A to FIG. 3H, so the same and similar descriptions in the two embodiments will not be repeated herein, and the main difference between the two embodiments will be described.

(51) Referring to FIG. 5A, the anti-reflection layer **170a** of this embodiment includes a yellow color filter layer **171Y** and a blue color filter layer **171B**. Specifically, by an ink-jet printing process (or a coating process and a photolithography process), the yellow color filter layer **171Y** is first formed on the insulating layer **160** and is disposed corresponding to the light-emitting region **102R**, the light-emitting region **102G**, and the non-light-emitting region **101**, but is not disposed corresponding to the light-emitting region **102B**. Then, the blue color filter layer **171B** is formed on the insulating layer **160** and the color filter layer **171Y** by the above method and is disposed corresponding to the light-emitting region **102B** and the non-light-emitting region **101**, but is not disposed corresponding to the light-emitting region **102R** and the light-emitting region **102G**. The stack of two color filter layers (i.e., the stack of the color filter layer **171Y** and the color filter layer **171B**) disposed on the non-light-emitting region **101** may have a light-shielding effect.

(52) FIG. 6A and FIG. 6B are respectively cross-sectional views showing a process of a method for manufacturing a display device according to some embodiments of the disclosure. FIG. 6A (or FIG. 6B) shows a step continuing FIG. 3D and replacing FIG. 3E to FIG. 3G. The same materials or methods may apply to the same or similar components in the embodiment of FIG. 6A (or the embodiment of FIG. 6B) and the embodiment of FIG. 3A to FIG. 3H, so the same and similar descriptions in the two embodiments will not be repeated herein, and the main difference between the two embodiments will be described.

(53) Referring to FIG. 2 and FIG. 6A, step S260 and step S260b are performed to form an anti-reflection layer **170b** on the insulating layer **160**, and the anti-reflection layer **170b** includes a polarizer **173**. In this embodiment, the anti-reflection layer **170b** may have a multi-layer structure, a continuous structure, or a patterned structure, and the anti-reflection layer **170b** further includes a first intermediate layer **172**, a second intermediate layer **174**, and a linear polarizer **175**. The first intermediate layer **172** is disposed on the insulating layer **160**, the second intermediate layer **174** is disposed on the polarizer **173**, and the linear polarizer **175** is disposed on the second intermediate layer **174**. The first intermediate layer **172** and the second intermediate layer **174** are respectively disposed on two sides of the polarizer **173** to protect the polarizer **173** and increase attachment between the polarizer **173** and the insulating layer **160** (or the linear polarizer **175**). In some embodiments, in the anti-reflection layer **170b**, the polarizer **173** may also be disposed directly on the insulating layer **160**, and the linear polarizer **175** may be disposed directly on the polarizer **173** without the first intermediate layer **172** and the second intermediate layer **174** (not shown). In this embodiment, the first intermediate layer **172**, the polarizer **173**, and the second intermediate layer **174** are sequentially formed directly on the insulating layer **160** by, for example, a coating process. The first intermediate layer **172**, the polarizer **173**, and the second intermediate layer **174** are all disposed corresponding to the light-emitting region **102R**, the light-emitting region **102G**, the light-emitting region **102B**, and the non-light-emitting region **101**. In this embodiment, the materials of the first intermediate layer **172** and the second intermediate layer **174** may include an organic material. In other embodiments of the disclosure, the first intermediate layer **172** and the second intermediate layer **174** may optionally not include an adhesive material, but the disclosure is not

limited thereto. The material of the polarizer **173** includes, for example, a quarter wave plate (QWP). The wavelength of the polarizer **173** is, for example, the average value of the red wavelength, the green wavelength, and the blue wavelength, and is, for example, 550 nanometers (nm), but the disclosure is not limited thereto. In addition, in this embodiment, a thickness **T10** of the first intermediate layer **172** is, for example, 0.01 μm to 0.1 μm but is not limited thereto. A thickness **T11** of the polarizer **173** is, for example, 1000 μm to 5000 μm but is not limited thereto. A thickness **T12** of the second intermediate layer **174** is, for example, 0.05 μm to 1 μm but is not limited thereto.

(54) Next, the linear polarizer **175** is formed directly on the second intermediate layer **174** by, for example, a coating process, and processes of aligning and baking (e.g., at 80° C. to 100° C., but the disclosure is not limited thereto) the linear polarizer **175** are performed, but the disclosure is not limited thereto. In some embodiments, the linear polarizer **175** may also be formed on the second intermediate layer **174** by, for example, a blade coating process, and the linear polarizer **175** may be aligned at the same time; then, the baking process of the linear polarizer **175** is performed. The alignment may include, for example, causing molecules in the linear polarizer **175** to be arranged in an orderly manner, so that the linear polarizer **175** can have a polarization function in a specific direction.

(55) Referring to FIG. 6B, an anti-reflection layer **170c** of this embodiment is similar to the anti-reflection layer **170b** of FIG. 6A, and the main difference between the two lies in that the anti-reflection layer **170c** of this embodiment may include a polarizer **173a**, a polarizer **173b**, and a polarizer **173c** designed for the light-emitting element **133** of different colors (the red light-emitting diode **133R**, the green light-emitting diode **133G**, and the blue light-emitting diode **133B**). The polarizer **173a** is disposed corresponding to at least the light-emitting region **102R** but is not disposed corresponding to the light-emitting region **102G** and the light-emitting region **102B**, the polarizer **173b** is disposed corresponding to at least the light-emitting region **102G** but is not disposed corresponding to the light-emitting region **102R** and light-emitting region **102B**, and the polarizer **173c** is disposed corresponding to at least the light-emitting region **102B** but is not disposed corresponding to the light-emitting region **102R** and the light-emitting region **102G**. In addition, the wavelength of the polarizer **173a** is, for example, $\frac{1}{4}$ of the red wavelength, the wavelength of the polarizer **173b** is, for example, $\frac{1}{4}$ of the green wavelength, and the wavelength of the polarizer **173c** is, for example, $\frac{1}{4}$ of the blue wavelength, but the disclosure is not limited thereto.

(56) FIG. 7 is a cross-sectional view showing a process of a method for manufacturing a display device according to another embodiment of the disclosure. FIG. 7 shows a step continuing FIG. 3D and replacing FIG. 3E to FIG. 3G. The same materials or methods may apply to the same or similar components in the embodiment of FIG. 7 and the embodiment of FIG. 3A to FIG. 3H, so the same and similar descriptions in the two embodiments will not be repeated herein, and the main difference between the two embodiments will be described.

(57) Referring to FIG. 2 and FIG. 7, step **S260** and step **S260c** are performed to form an anti-reflection layer **170e** on the insulating layer **160**. The anti-reflection layer **170e** includes a wire-grid polarizer (WGP) **176**, and the wire-grid polarizer **176** may include a patterned wire-grid polarizer. In this embodiment, for example, the following two methods are adopted to form the wire-grid polarizer **176** directly on the insulating layer **160**, but the disclosure is not limited thereto. In the first method, first, the wire-grid polarizer **176** is deposited on the insulating layer **160** to directly cover the insulating layer **160**; next, a plurality of openings **176a** are formed by a photolithography process to expose the insulating layer **160**; and then, a blackening treatment (e.g., oxidation) is performed to reduce reflection caused by the wire-grid polarizer **176**. In the second method, the wire-grid polarizer **176** is formed directly on the insulating layer **160** by nanoimprint lithography, so that the wire-grid polarizer **176** directly covers the insulating layer **160** and has a plurality of openings **176a** exposing the insulating layer **160**; and then, a blackening treatment (e.g., oxidation)

is performed to reduce reflection caused by the wire-grid polarizer **176**. Next, after the wire-grid polarizer **176** is formed on the insulating layer **160**, an insulating layer **177** is formed on the wire-grid polarizer **176** and in the openings **176a** of the wire-grid polarizer **176**. In this embodiment, the material of the wire-grid polarizer **176** may include aluminum, silver, platinum, gold, or a metal compound but is not limited thereto.

(58) FIG. **8A** and FIG. **8B** are schematic partial cross-sectional views of a display device according to some embodiments of the disclosure. Referring to FIG. **1** and FIG. **8A** (or FIG. **8B**) at the same time, a display device **100b** (or a display device **100c**) of this embodiment is substantially similar to the display device **100** of FIG. **1**, so the same and similar components in the two embodiments will not be repeatedly described herein. The main difference between the display device **100b** (or the display device **100c**) of this embodiment and the display device **100** lies in that the display device **100b** (or the display device **100c**) further includes an insulating layer **161** (or an insulating layer **162**).

(59) Specifically, referring to FIG. **8A**, the insulating layer **161** is disposed between the insulating layer **160** and the touch sensing unit **150**. The insulating layer **161** has a plurality of recesses **161a**, and the recesses **161a** do not penetrate the insulating layer **161**. The recess **161a** may be disposed corresponding to the non-light-emitting region **101**, but is not disposed corresponding to the light-emitting regions **102R**, **102G**, and **102B**, so that the recess **161a** can scatter external light to improve the anti-reflection effect.

(60) In addition, in other embodiments, other morphology designs may also be adopted to replace the recess to achieve the anti-reflection effect. As shown in FIG. **8B**, the insulating layer **162** of the display device **100c** is disposed between the insulating layer **160** and the touch sensing unit **150** and has a plurality of rough surfaces **162a**. The rough surface **162a** may be disposed corresponding to the non-light-emitting region **101**, but is not disposed corresponding to the light-emitting regions **102R**, **102G**, and **102B**, so that the rough surface **162a** can scatter external light to improve the anti-reflection effect.

(61) FIG. **9** is a schematic top view of a display device according to another embodiment of the disclosure. FIG. **10A** to FIG. **10E** are cross-sectional views showing a process of a method for manufacturing the display device of FIG. **9** taken along section line A-A'. FIG. **11A** to FIG. **11C** are cross-sectional views showing a process of a method for manufacturing the display device of FIG. **9** taken along section line B-B'. For clarity of the drawings and convenience of illustration, FIG. **9** omits several components in a display device **100d**. In addition, the anti-reflection layer **170** and/or the cover layer **180** of island regions **103**, **103a**, and **103b** in the display device **100d** may be designed according to any of the above embodiments. The description below is based on a design similar to that shown in FIG. **3H**, but the disclosure is not limited thereto.

(62) Referring to FIG. **9**, the display device **100d** includes a plurality of island regions **103**, **103a**, and **103b**, a plurality of bridge regions **104**, and a plurality of opening regions **105**. The island regions **103**, **103a**, and **103b** are separate from each other and each includes a plurality of light-emitting units **130**. The bridge regions **104** are respectively disposed between two adjacent island regions (e.g., between the island region **103** and the island region **103a**, or between the island region **103a** and the island region **103b**), and are configured to connect two adjacent island regions (e.g., the island region **103** and the island region **103a**, or the island region **103a** and the island region **103b**). The opening regions **105** have holes **190** which penetrate the display device **100d**. The opening regions **105** are respectively disposed between two adjacent island regions (e.g., between the island region **103** and the island region **103b**) to isolate two adjacent island regions (e.g., the island region **103** and the island region **103b**) by the hole **190**. In this embodiment, with the design of the bridge regions **104** and the opening regions **105**, the island regions **103**, **103a**, and **103b** can be separate from each other, and the display device **100d** of this embodiment can be stretchable. The stretchable display device **100d** may be fit on an irregular surface (e.g., a spherical surface, a curved surface, an undulating surface, etc.) or may be stretched and enlarged for use. The

expression “stretchable” mentioned in this disclosure may mean, for example, that the size of the display device may be stretched from 5 inches to 6 inches, or the distance between the island region **103** and the island region **103a** or between the island region **103a** and the island region **103b** of the display device may be changed by any operation, but the disclosure is not limited thereto.

(63) Next, since the embodiment shown in FIG. **10A** to FIG. **10E** is similar to the embodiment shown in FIG. **3A** to FIG. **3H**, the same materials or methods may apply to the same or similar components and the same and similar descriptions in the two embodiments will not be repeated herein. The difference between the embodiment shown in FIG. **10A** to FIG. **10E** and the embodiment shown in FIG. **3A** to FIG. **3H** lies in that the method for manufacturing the display device **100d** of this embodiment further includes forming the bridge region **104** between the island region **103** and the island region **103a** adjacent to each other.

(64) Specifically, referring to FIG. **10A** and FIG. **10B** first, sequentially, a pixel circuit **120** is formed on the substrate **110**, a light-emitting unit **130** is formed on the pixel circuit **120**, an encapsulation layer **140** is formed on the light-emitting unit **130**, and a touch sensing unit **150** is formed on the light-emitting unit **130**. A transistor **121** in the pixel circuit **120** and the light-emitting unit **130** are both disposed in the island regions **103** and **103a**, but are not disposed in the bridge region **104**. In the bridge region **104**, a metal layer GE1 may be disposed on a gate insulating layer GI, and a metal layer SD3 may be disposed on an insulating layer **122**. The light-emitting unit **130** includes a red light-emitting diode **133R**, a green light-emitting diode **133G**, and a blue light-emitting diode **133B**. The touch sensing unit **150** is disposed on the encapsulation layer **140** in the island regions **103** and **103a** and the bridge region **104**, so that the touch sensing unit **150** in the island regions **103** and **103a** and the touch sensing unit **150** in the bridge region **104** may be located on the same horizontal plane and may be easier to manufacture.

(65) Next, referring to FIG. **10C**, an anti-reflection layer **170** is formed on the insulating layer **160**, and the anti-reflection layer **170** includes a red color filter layer **171R**, a green color filter layer **171G**, and a blue color filter layer **171B**. Specifically, the red color filter layer **171R** is disposed on the insulating layer **160** and is disposed corresponding to the light-emitting region **102R** (i.e., the light-emitting region of the red light-emitting diode **133R**) and the non-light-emitting region **101** in the island regions **103** and **103a** and the bridge region **104**. The green color filter layer **171G** is disposed on the insulating layer **160** and the color filter layer **171R** and is disposed corresponding to the light-emitting region **102G** and the non-light-emitting region **101** in the island regions **103** and **103a** and the bridge region **104**. The blue color filter layer **171B** is disposed on the insulating layer **160** and the color filter layer **171G** and is disposed corresponding to the light-emitting region **102B** and the non-light-emitting region **101** in the island regions **103** and **103a** and the bridge region **104**.

(66) Then, referring to FIG. **10D** and FIG. **10E**, a groove **191** is formed at the junction between the island regions **103** and **103a** and the bridge region **104**, and a cover layer **180** is formed on the anti-reflection layer **170** and in the groove **191**. In this embodiment, the groove **191** may be formed by, for example, etching, laser drilling, or a combination of the above but is not limited thereto. The groove **191** may expose part of the insulating layer **123** in the pixel circuit **120**, and the groove **191** does not penetrate the substrate **110**, so stress can be released accordingly. The cover layer **180** may cover the anti-reflection layer **170** and a sidewall **106** of the display device **100d**. So far, the island regions **103** and **103a** and the bridge region **104** in the display device **100d** of this embodiment have been completed.

(67) In addition, since the embodiment shown in FIG. **11A** to FIG. **11C** is similar to the embodiment shown in FIG. **3A** to FIG. **3H**, the same materials or methods may apply to the same or similar components and the same and similar descriptions in the two embodiments will not be repeated herein. The difference between the embodiment shown in FIG. **11A** to FIG. **11C** and the embodiment shown in FIG. **3A** to FIG. **3H** lies in that the method for manufacturing the display device **100d** of this embodiment further includes forming an opening region **105** between the island

region **103** and the island region **103b** adjacent to each other.

(68) Specifically, FIG. **11A** to FIG. **11C** may be regarded as steps continuing FIG. **10C** and replacing FIG. **10D** to FIG. **10E**, and the manufacturing continues with the island region **103b** and the opening region **105** in FIG. **11A** to FIG. **11C** respectively replacing the island region **103a** and the bridge region **104** in FIG. **10A** to FIG. **10C**. Referring to FIG. **11A** first, the anti-reflection layer **170**, the insulating layer **160**, the touch sensing unit **150**, the encapsulation layer **140**, and the light-emitting unit **130** of the opening region **105** are etched and/or drilled to form a hole **192**. The hole **192** may expose part of the insulating layer **123** in the pixel circuit **120**.

(69) Next, referring to FIG. **11B**, the pixel circuit **120** and the substrate **110** of the opening region **105** are etched and/or drilled to form a hole **193**. The hole **192** is connected to the hole **193** to form a hole **190** penetrating the display device **100d**. A sidewall **192a** of the hole **192** and a sidewall **193a** of the hole **193** may be rough surfaces. It is possible that the sidewall **192a** of the hole **192** and the sidewall **193a** of the hole **193** are not flush with each other. In addition, although the hole **192** and the hole **193** in this embodiment are formed in two steps, in some embodiments, the hole **192** and the hole **193** may also be formed simultaneously in one step. In this embodiment, the hole **192** and the hole **193** may be formed by, for example, etching, laser drilling, or a combination of the above but are not limited thereto.

(70) Then, referring to FIG. **11C**, a cover layer **180** is formed on the anti-reflection layer **170**, in the hole **192**, and in the hole **193**. The cover layer **180** may cover the anti-reflection layer **170**, the sidewall **106** of the display device **100d**, the sidewall **192a** of the hole **192**, and the sidewall **193a** of the hole **193**. So far, the island regions **103** and **103b** and the opening region **105** in the display device **100d** of this embodiment have been completed.

(71) FIG. **12A** to FIG. **12C** are cross-sectional views showing a process of a method for manufacturing the display device of FIG. **9** taken along section line A-A' according to another embodiment. Since the embodiment shown in FIG. **12A** to FIG. **12C** is similar to the embodiment shown in FIG. **10A** to FIG. **10E**, the same materials or methods may apply to the same or similar components and the same and similar descriptions in the two embodiments will not be repeated herein. The difference between the embodiment shown in FIG. **12A** to FIG. **12C** and the embodiment shown in FIG. **10A** to FIG. **10E** lies in that, in the method for manufacturing the bridge region **104** of a display device **100e** of this embodiment, first, a hole **191'** is formed in the bridge region **104**, and then a touch sensing unit **150'** and an anti-reflection layer **170'** are formed in the island regions **103** and **103a** and the bridge region **104**.

(72) Specifically, referring to FIG. **12A** which shows a step continuing FIG. **10A**, the hole **191'** is formed in the bridge region **104** so that the hole **191'** can expose part of the insulating layer **123** in the pixel circuit **120**. Then, referring to FIG. **12B**, the touch sensing unit **150'** is formed on the encapsulation layer **140** in the island regions **103** and **103a** and in the hole **191'** in the bridge region **104**. Since the touch sensing unit **150'** in the bridge region **104** is disposed on the insulating layer **123** in the hole **191'**, the touch sensing unit **150'** in the bridge region **104** may be closer to the pixel circuit **120** and the light-emitting unit **130**, so that the stress of the bridge region **104** can be more concentrated and easier to adjust or the bridge region **104** may have a better stretching effect. Then, referring to FIG. **12C**, the anti-reflection layer **170'** is formed on the insulating layer **160**, and a cover layer **180'** is formed on the anti-reflection layer **170'**. The anti-reflection layer **170'** may be disposed on the insulating layer **160** in the island regions **103** and **103a** and on the insulating layer **160** in the hole **191'** in the bridge region **104**. The cover layer **180'** may be disposed on the anti-reflection layer **170'** in the island regions **103** and **103a** and on the anti-reflection layer **170'** in the hole **191'** in the bridge region **104**. So far, the island regions **103** and **103a** and the bridge region **104** in the display device **100e** of this embodiment have been completed.

(73) In addition, the opening region of the display device **100e** in this embodiment may also be manufactured by a method different from that in FIG. **11A** to FIG. **11C**. The main difference between the method for manufacturing the opening region of this embodiment and the

manufacturing method of FIG. 11A to FIG. 11C lies in that, in the method for manufacturing the opening region of the display device **100e** of this embodiment, first, a hole which exposes part of the insulating layer **123** in the pixel circuit **120** is formed in the opening region, which is similar to the step shown in FIG. 12A (when the bridge region **104** in FIG. 12A is regarded as the opening region, and the island region **103a** is regarded as the island region **103b**). Next, a touch sensing unit **150'** and an anti-reflection layer **170'** are formed in the island regions **103** and **103b**, which similar to the step shown in FIG. 11A. Then, a hole penetrating the display device **100e** is formed in the opening region, which is similar to the step shown in FIG. 11B. Then, a cover layer **180** is formed on the anti-reflection layer **170'**, which is similar to the step shown in FIG. 11C. So far, the island regions **103**, **103b** and the opening region in the display device **100e** of this embodiment have been completed.

(74) FIG. 13 is a flowchart of a method for manufacturing a display device according to another embodiment of the disclosure. FIG. 14A to FIG. 14D are cross-sectional views showing a process of a method for manufacturing a display device according to another embodiment of the disclosure. Since the embodiment shown in FIG. 2 (or FIG. 3A to FIG. 3H) is similar to the embodiment shown in FIG. 13 (or FIG. 14A to FIG. 14D), the same materials or methods may apply to the same or similar components and the same and similar descriptions in the two embodiments will not be repeated herein. The difference between the embodiment shown in FIG. 2 (or FIG. 3A to FIG. 3H) and the embodiment shown in FIG. 13 (or FIG. 14A to FIG. 14D) lies in that, in the method for manufacturing a display device **100f** of this embodiment, a light-emitting element **133f** is formed on the pixel circuit **120** by transfer and bonding.

(75) Specifically, referring to FIG. 13 and FIG. 14A at the same time, in the method for manufacturing the display device **100f** of this embodiment, first, step S300 and step S310 are performed to provide a substrate **110** and form a pixel circuit **120** and a touch sensing unit **150f** on the substrate **110**. Specifically, a buffer layer **111** is further included between the pixel circuit **120** and the substrate **110**. The touch sensing unit **150f** may include a first metal layer **151f** and an insulating layer **152f**. The first metal layer **151f** and the insulating layer **152f** are adjacently disposed on the insulating layer **123** of the pixel circuit **120**. The first metal layer **151f** is disposed corresponding to the non-light-emitting region **101**, but is not disposed corresponding to the light-emitting regions **102R**, **102G**, and **102B**.

(76) Next, referring to FIG. 13, FIG. 14A, and FIG. 14B, step S320 is performed to form a light-emitting unit **130f** on the touch sensing unit **150f**. Specifically, the light-emitting unit **130f** includes a first electrode **131f**, an insulating layer **132f**, a light-emitting element **133f**, and a second electrode **134f**. The first electrode **131f** and the second electrode **134f** are adjacently disposed on the insulating layer **152f**. The insulating layer **132f** is disposed on the first electrode **131f** and the second electrode **134f**. The insulating layer **132f** has an opening **132f1** to expose part of the first electrode **131f** and part of the second electrode **134f**. The light-emitting element **133f** which has been completed is transferred into the opening **132f1** to bond to the first electrode **131f** and the second electrode **134f** in the opening **132f1** and electrically connect to the transistor **121** via the first electrode **131f**. In this embodiment, the light-emitting element **133f** may be, for example, a flip-chip micro light-emitting diode but is not limited thereto. The light-emitting element **133f** may include a red light-emitting diode, a green light-emitting diode, and a blue light-emitting diode but is not limited thereto. In some embodiments, the light-emitting element **133f** may also be a blue light-emitting diode.

(77) Then, referring to FIG. 13, FIG. 14C, and FIG. 14D, step S330, step S340, and step S350 are performed to form an encapsulation layer **140f** on the light-emitting unit **130f**, form an anti-reflection layer **170f** on the encapsulation layer **140f**, and form a cover layer **180** on the anti-reflection layer **170f**. The encapsulation layer **140f** may be disposed in the opening **132f1** and may cover the light-emitting element **133f** and the second electrode **134f**. The anti-reflection layer **170f** may include a color filter layer, a polarizer, or a wire-grid polarizer. So far, the display device **100f**

of this embodiment has been completed.

(78) FIG. 15 is a flowchart of a method for manufacturing a display device according to another embodiment of the disclosure. FIG. 16A to FIG. 16D are cross-sectional views showing a process of a method for manufacturing a display device according to another embodiment of the disclosure. Since the embodiment shown in FIG. 13 (or FIG. 14A to FIG. 14D) is similar to the embodiment shown in FIG. 15 (or FIG. 16A to FIG. 16D), the same materials or methods may apply to the same or similar components and the same and similar descriptions in the two embodiments will not be repeated herein. The difference between the embodiment shown in FIG. 13 (or FIG. 14A to FIG. 14D) and the embodiment shown in FIG. 15 (or FIG. 16A to FIG. 16D) lies in that, the method for manufacturing a display device 100g of this embodiment further includes forming a light conversion layer 200 on an encapsulation layer 140g.

(79) Specifically, referring to FIG. 15 and FIG. 16A at the same time, in the method for manufacturing the display device 100g of this embodiment, first, steps S300 and S310 are performed to provide a substrate 110 and form a pixel circuit 120 and a touch sensing unit 150f on the substrate 110. Specifically, a buffer layer 111 is further included between the pixel circuit 120 and the substrate 110. The touch sensing unit 150f may include a first metal layer 151f and an insulating layer 152f. The first metal layer 151f and the insulating layer 152f are adjacently disposed on an insulating layer 123 of the pixel circuit 120. The first metal layer 151f is disposed corresponding to the non-light-emitting region 101 but is not disposed corresponding to the light-emitting regions 102R, 102G, and 102B.

(80) Next, referring to FIG. 15, FIG. 16A, and FIG. 16B, step S320 is performed to form a light-emitting unit 130g on the touch sensing unit 150f. Specifically, the light-emitting unit 130g includes a first electrode 131g, an insulating layer 132g, a light-emitting element 133g, a second electrode 134g, and an insulating layer 135g. The first electrode 131g is disposed on the insulating layer 152f. The insulating layer 132g is disposed on the first electrode 131g. The insulating layer 132g has an opening 132g1 to expose part of the first electrode 131g. The light-emitting element 133g which has been completed is transferred into the opening 132g1 to bond to the first electrode 131g in the opening 132g1 and electrically connect to the transistor 121 via the first electrode 131g. The insulating layer 135g is disposed on the insulating layer 132g and in the opening 132g1 to cover the light-emitting element 133g. The second electrode 134g is disposed on the insulating layer 135g and is electrically connected to the light-emitting element 133g. In this embodiment, the light-emitting element 133g may be, for example, a vertical micro light-emitting diode or a vertical nano light-emitting diode (nano-LED) but is not limited thereto. The light-emitting element 133g may include a blue light-emitting diode but is not limited thereto.

(81) Then, referring to FIG. 15 and FIG. 16C, step S330 and step S331 are performed to form an encapsulation layer 140g on the light-emitting unit 130g and form a light conversion layer 200 on the encapsulation layer 140g. The encapsulation layer 140g may cover the insulating layer 135g and the second electrode 134g. The light conversion layer 200 may include a red light conversion layer 200R and a green light conversion layer 200G respectively disposed corresponding to one of the light-emitting elements 133g. In this embodiment, the light conversion layer 200 may include, for example, quantum dots but is not limited thereto.

(82) Then, referring to FIG. 15 and FIG. 16D, step S341 and step S350 are performed to form an anti-reflection layer 170f on the light conversion layer 200 and form a cover layer 180 on the anti-reflection layer 170f. Before the anti-reflection layer 170f is formed on the light conversion layer 200, an insulating layer 210 may be further formed on the light conversion layer 200, so that the insulating layer 210 can cover the light conversion layer 200 and the insulating layer 135g. The anti-reflection layer 170f may include a color filter layer, a polarizer, or a wire-grid polarizer. So far, the display device 100g of this embodiment has been completed.

(83) In summary of the above, in the method for manufacturing the display device of the embodiments of the disclosure, compared to a method of attaching an anti-reflection layer

externally onto the insulating layer, the method of forming the anti-reflection layer on the insulating layer in the embodiments of the disclosure can reduce the overall thickness, reduce the bending stress, improve the reliability, or increase the service life. For example, when the display device is a foldable display device or a stretchable display device, and when the display device is to be bent, rolled, or stretched, compared to the method of attaching an anti-reflection layer externally onto the insulating layer, the method of forming the anti-reflection layer on the insulating layer in the embodiments of the disclosure can reduce the bending stress to suppress separation of the anti-reflection layer, which thereby improves the reliability of the display device or increases the service life of the display device.

(84) Finally, it should be noted that the foregoing embodiments are merely used for describing the technical solutions of the disclosure, but are not intended to limit the disclosure. Although the disclosure has been described in detail with reference to the foregoing embodiments, a person of ordinary skill in the art should understand that modifications may still be made to the technical solutions in the foregoing embodiments, or equivalent replacements may be made to part or all of the technical features; and these modifications or replacements will not cause the essence of corresponding technical solutions to depart from the scope of the technical solutions in the embodiments of the disclosure.

Claims

1. A method for manufacturing a display device, comprising: providing a substrate; forming a pixel circuit on the substrate; forming a light-emitting unit on the pixel circuit; forming a touch sensing unit on the light-emitting unit; and forming an anti-reflection layer on the touch sensing unit, wherein a portion of the anti-reflection layer comprises a stack having at least two of a red color filter layer, a green color filter layer and a blue color filter layer, and the stack is disposed on the touch sensing unit by a coating process, a photolithography process or an ink-jet printing process, wherein the touch sensing unit is disposed between the anti-reflection layer and the substrate, wherein in a cross-section view, a top surface of the anti-reflection layer has a first portion corresponding to the blue color filter layer, a second portion corresponding to the red color filter layer and a third portion corresponding to the green color filter layer, and a projection length of the first portion on the substrate is greater than a projection length of the second portion on the substrate or a projection length of third portion on the substrate.
2. The method as claimed in claim 1, wherein the anti-reflection layer is formed directly on an insulating layer.
3. The method as claimed in claim 1, further comprising: forming a hole penetrating the display device.
4. The method as claimed in claim 1, wherein the display device is stretchable.
5. The method as claimed in claim 1, wherein the substrate has a light-emitting region and a non-light-emitting region, and the light-emitting unit comprises a light-emitting element, wherein the light-emitting element overlaps with the light-emitting region, and the anti-reflective layer overlaps with the non-light-emitting region.
6. The method as claimed in claim 1, further comprising: forming a plurality of holes penetrating the display device; wherein the display device comprises: a plurality of island regions comprising a plurality of light-emitting units; a plurality of bridge regions respectively connected two adjacent island regions; and a plurality of opening regions respectively disposed between two adjacent island regions and comprising the plurality of holes; wherein the plurality of island regions are separated from each other by the plurality of holes.
7. The method as claimed in claim 1, wherein the display device comprises: a plurality of island regions separated from each other and comprising a plurality of light-emitting units; a plurality of bridge regions respectively connected two adjacent island regions; and a plurality of opening

regions respectively disposed between two adjacent island regions for isolating the plurality of island regions; wherein the display device is stretchable when a distance between two adjacent island regions is changed.
